CS F320 Foundations of Data Science

Assignment 1: Polynomial, Ridge, Lasso and Bayesian Regression

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Implementation

Everything is written from scratch in Python 3.12 with NumPy for linear algebra and Matplotlib for plots; no fitting library is used. The code is at `https://github.com/Chaitanya-Keyal/polynomial-regression`: `src/p1.py` to `src/p4.py` hold one problem each, `src/common/` the shared bases and solvers, and `python -m src.run` regenerates every figure and every number in this report from the six provided CSV files, which are loaded as given. $\sin(2\pi x)$ appears only in plots. The random seed for the per-epoch shuffling in P1 is 42 (a fresh `np.random.default_rng(42)` per run), so every result is reproducible to the last digit. No explicit matrix inverse is formed: least squares uses the SVD-based `np.linalg.lstsq`, and ridge and the Bayesian posterior use `np.linalg.solve`. RMSE is $\sqrt{\text{mean squared error}}$ throughout.

1 P1: Closed form vs. the gradient descent family

Method. The degree-d model $y(x, \mathbf{w}) = \sum_{j=0}^{d} w_j x^j$ has design matrix $\mathbf{\Phi} \in \mathbb{R}^{N\times(d+1)}$ with rows $[1, x_n, \ldots, x_n^d]$ and objective $E(\mathbf{w}) = \frac{1}{2}\|\mathbf{\Phi}\mathbf{w} - \mathbf{t}\|^2$. The closed-form minimiser solves $\mathbf{\Phi}^\top\mathbf{\Phi}\mathbf{w} = \mathbf{\Phi}^\top\mathbf{t}$, but for $d = 9$ we have $\text{cond}(\mathbf{\Phi}) = 1.52 \times 10^7$, and forming $\mathbf{\Phi}^\top\mathbf{\Phi}$ would square that to about 10^{14}. We therefore solve the least squares problem directly with `np.linalg.lstsq`, which works through the SVD of $\mathbf{\Phi}$.

For gradient descent we take the thin QR decomposition $\mathbf{\Phi} = QR$, set $\boldsymbol{\alpha} = R\mathbf{w}$ and minimise $E(\boldsymbol{\alpha}) = \frac{1}{2}\|Q\boldsymbol{\alpha} - \mathbf{t}\|^2$ from $\boldsymbol{\alpha} = \mathbf{0}$, then recover $\mathbf{w} = R^{-1}\boldsymbol{\alpha}$ by a solve with R ($\text{cond}(Q) = 1.000$). With $\mathbf{q}_n^\top$ the n-th row of Q and $r_n = \mathbf{q}_n^\top\boldsymbol{\alpha} - t_n$ its residual, one epoch is

- **Batch GD**: one step $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta\, Q^\top(Q\boldsymbol{\alpha} - \mathbf{t})$, the full sum, not divided by N;
- **Stochastic GD**: shuffle, then for each n in turn $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \eta\, r_n\mathbf{q}_n$ ($N = 10$ updates);
- **Mini-batch GD**: shuffle, then for each batch B of the shuffled order (sizes 3, 3, 3, 1) $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} - \frac{\eta}{|B|}\sum_{n\in B} r_n\mathbf{q}_n$ (4 updates).

Closed-form fits. Figure 1 shows the least-squares fits for $d = 1, \ldots, 9$ and Table 1 their training error. The models are nested, so the training error can only fall as d grows; at $d = 9$ there are as many weights as points and the polynomial interpolates the data ($E \approx 0$). Degrees 1 and 2 underfit, degrees 3 to 6 follow $\sin(2\pi x)$ closely, and from $d = 7$ on the curve bends between the points to chase the noise, most visibly near the ends of $[0, 1]$.

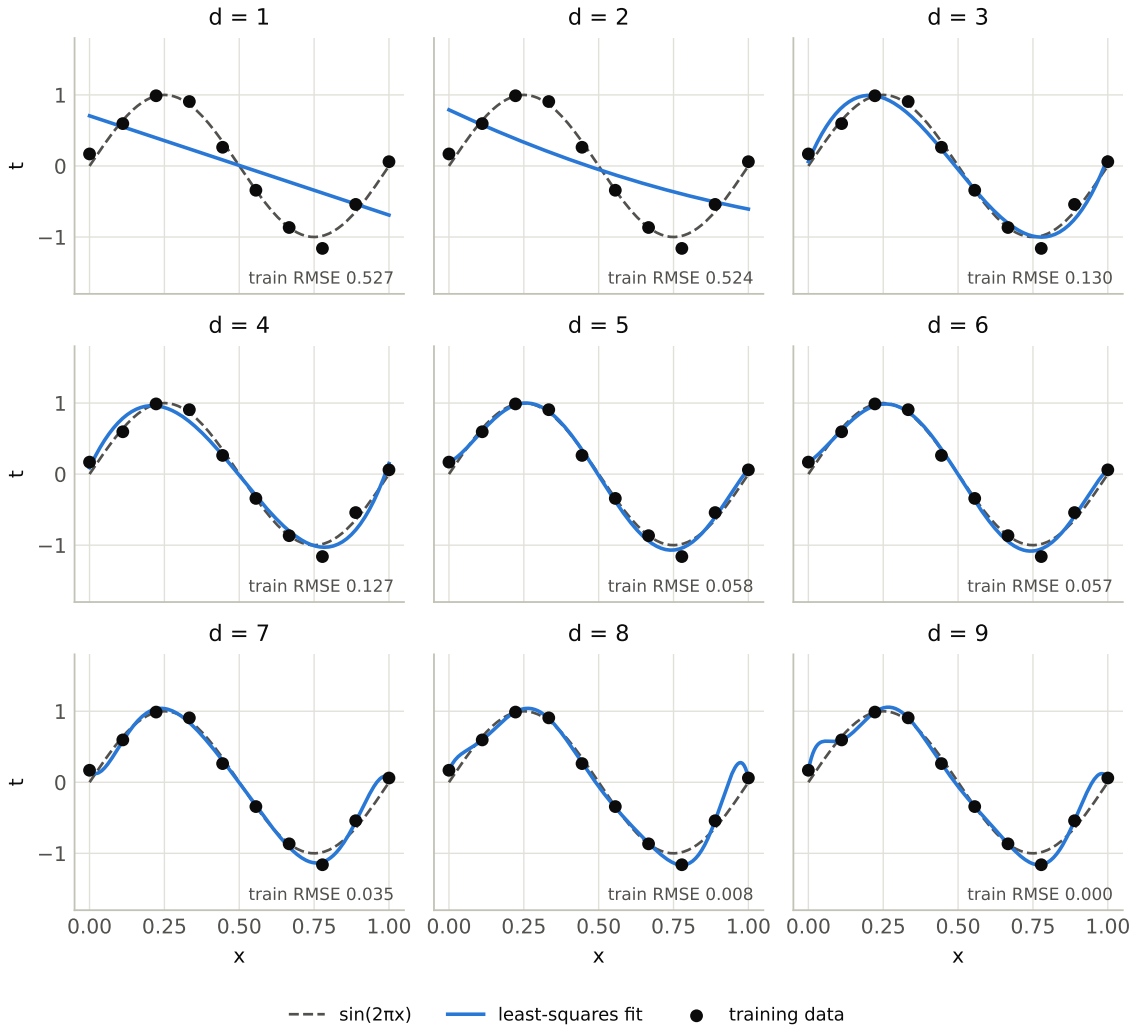


Figure 1: Least-squares polynomial fits of degree 1 to 9 to the $N = 10$ training points, with the true function $\sin(2\pi x)$.

d	training $E(\mathbf{w})$	training RMSE
1	1.39	0.527
2	1.37	0.524
3	0.0844	0.130
4	0.0808	0.127
5	0.0166	0.0575
6	0.0161	0.0568
7	0.00627	0.0354
8	3.09×10^{-4}	0.00787
9	6.61×10^{-23}	3.64×10^{-12}

Table 1: Training error of the closed-form fits.

Loss curves. Figure 2 shows E per epoch for the three variants at $d = 9$ and $\eta = 0.1$ over all 2000 epochs of the verification run: every curve has flattened at the floating-point floor (Batch near 10^{-30}, SGD and Mini-batch near 10^{-29}, since their 10 and 4 rounded updates per epoch leave more rounding). Batch and Stochastic GD reach $E < 10^{-12}$ after 136 and 136 epochs, Mini-batch GD after 353; over the first 100 epochs E shrinks per epoch by an average factor of 0.8100, 0.8100 and 0.9242.

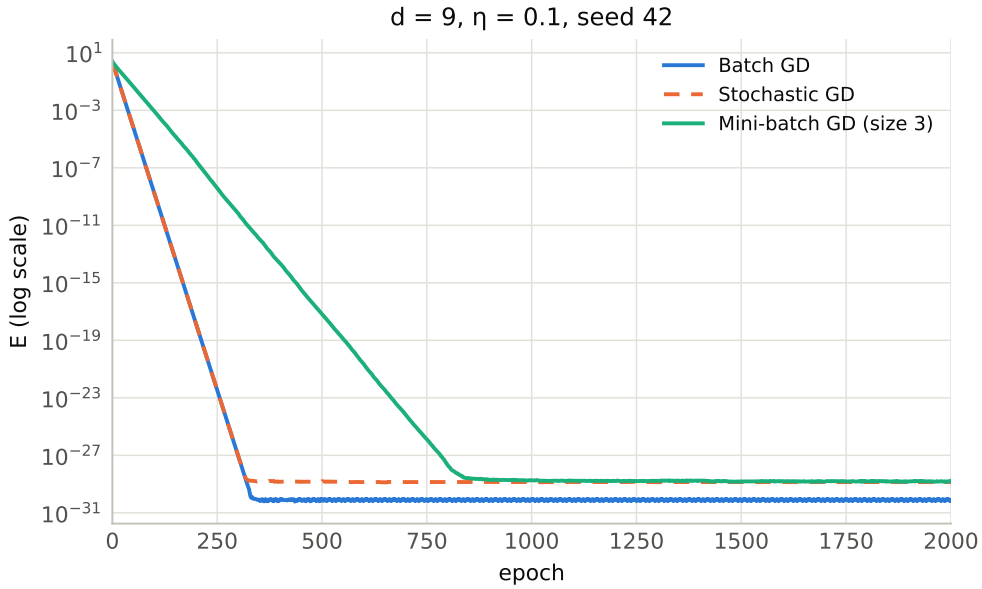


Figure 2: E vs. epoch for Batch, Stochastic and Mini-batch GD ($d = 9$, $\eta = 0.1$, seed 42). The SGD curve (dashed) lies on the Batch curve until both reach the floating-point floor; below it only rounding separates them.

Learning-rate study. In the QR coordinates the Hessian of E is $Q^\top Q = I$, so every eigenvalue is 1 and Batch GD gives $\boldsymbol{\alpha}_k - \boldsymbol{\alpha}^\star = (1-\eta)^k(\boldsymbol{\alpha}_0 - \boldsymbol{\alpha}^\star)$ and $E_k = (1-\eta)^{2k}E_0$ with $E_0 = \frac{1}{2}\|\mathbf{t}\|^2 = 2.381$. GD converges exactly when $|1-\eta| < 1$, i.e. $0 < \eta < 2$, fastest at $\eta = 1$ (one epoch). Figure 3 shows the three regimes over 100 epochs:

- $\eta = 0.01$ (too small): E is multiplied by only $0.99^2 \approx 0.98$ per epoch and is still 0.3190 after 100 epochs;
- $\eta = 0.5$ (well chosen): E falls by a factor of 4 per epoch and reaches the floating-point floor within about 50 epochs ($E_{100} = 5.106 \times 10^{-32}$);
- $\eta = 2.1$ (too large): $|1-\eta| = 1.1$, so the error flips sign and grows by 1.1 every epoch. After 100 epochs $E = 4.521 \times 10^8 = 1.1^{200}E_0$, and $\boldsymbol{\alpha}_{100} = (1 - 1.1^{100})\boldsymbol{\alpha}^\star$, so the weight vector is -13780 times the closed-form solution (Table 2).

At exactly $\eta = 2$ the factor is -1: $\boldsymbol{\alpha}$ alternates between $\mathbf{0}$ and $2\boldsymbol{\alpha}^\star$, each step overshooting the minimum by exactly as much as it started from it, and E stays at E_0 forever. It neither converges nor diverges.

j	$w_{\text{closed},j}$	$w_{100,j}$ at $\eta = 2.1$
0	0.169	-2329
1	21.3	-2.93×10^5
2	-414	5.70×10^6
3	3788	-5.22×10^7
4	-17589	2.42×10^8
5	45606	-6.28×10^8
6	-69409	9.56×10^8
7	61610	-8.49×10^8
8	-29469	4.06×10^8
9	5856	-8.07×10^7

Table 2: The closed-form weights for $d = 9$, and the weights Batch GD reaches after 100 epochs with $\eta = 2.1$: the same vector scaled by $1 - 1.1^{100}$.

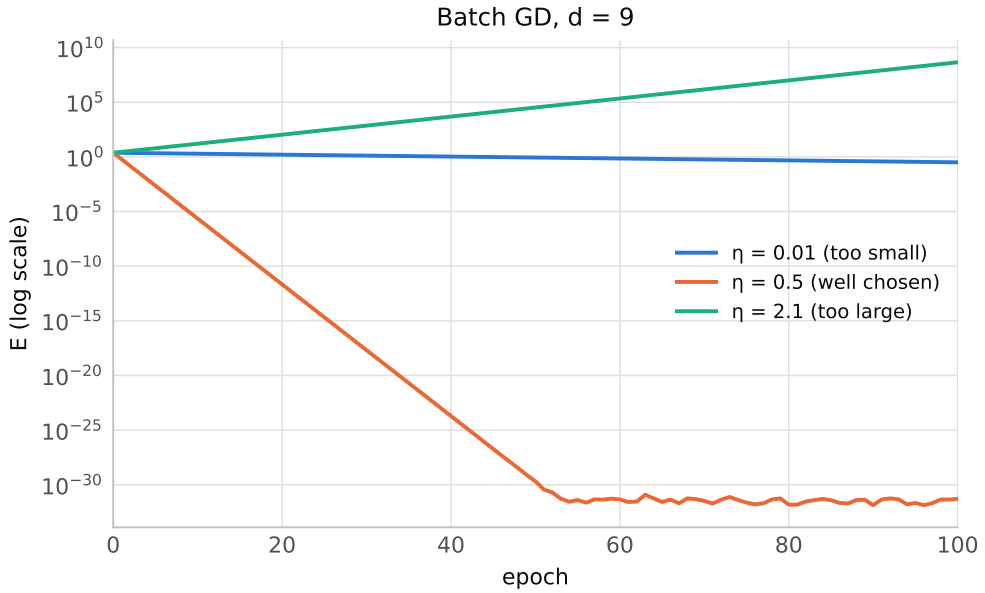


Figure 3: Batch GD ($d = 9$, QR-reparametrised) for a learning rate that is too small, well chosen and too large.

Verification. After 2000 epochs at $\eta = 0.1$:

variant	$\|\mathbf{w}_{\text{GD}} - \mathbf{w}_{\text{closed}}\|_2$	relative error	final E
Batch GD	4.06×10^{-6}	3.72×10^{-11}	7.0×10^{-31}
Stochastic GD	4.06×10^{-6}	3.72×10^{-11}	1.4×10^{-29}
Mini-batch GD	4.06×10^{-6}	3.72×10^{-11}	1.6×10^{-29}

Table 3: GD against the closed form, $d = 9$; $\|\mathbf{w}_{\text{closed}}\|_2 = 1.09 \times 10^5$.

The weights are large (Table 2), so an absolute error of this size is a relative error of only 3.72×10^{-11}: all three variants agree with the closed form to the precision the problem allows. The error is not optimisation error, since GD converges to the QR solution $R^{-1}Q^\top \mathbf{t}$ to within rounding; what remains is the gap between two direct solutions of the same system (QR, and the SVD inside `lstsq`), well inside the relative error of about $\text{cond}(\mathbf{\Phi})\,\epsilon_{\text{mach}} \approx 3 \times 10^{-9}$ that rounding allows.

Discussion. In Figure 2 the SGD curve lies on the Batch curve epoch for epoch (relative gap below 6.8×10^{-11} while $E > 10^{-10}$) until both reach the rounding floor, while Mini-batch GD is clearly slower. (i) With $N = d + 1 = 10$ the thin Q is square and orthogonal, so its *rows* $\mathbf{q}_n$ are orthonormal: an SGD step on point n changes residual r_m by $-\eta r_n \mathbf{q}_m^\top \mathbf{q}_n$, which multiplies r_n by $(1 - \eta)$ and leaves every other residual untouched, so one pass in any order multiplies every residual by $(1 - \eta)$ once, exactly like one Batch step $\mathbf{r} \leftarrow (I - \eta QQ^\top)\mathbf{r} = (1 - \eta)\mathbf{r}$. (ii) Mini-batch GD *averages* over its batch, so the nine points in batches of three shrink only by $(1 - \eta/3)$ per epoch (the singleton by $1 - \eta$): E falls by about 0.9242 per epoch instead of 0.81 and needs about 2.6 times as many epochs (353 vs. 136).

2 P2: Ridge regression as a MAP estimate

MAP $\Rightarrow$ ridge. By Bayes' rule $p(\mathbf{w} \mid \mathbf{t}) \propto p(\mathbf{t} \mid \mathbf{w})\, p(\mathbf{w})$ with $p(\mathbf{t} \mid \mathbf{w}) \propto \exp(-\frac{\beta}{2}\|\mathbf{\Phi w} - \mathbf{t}\|^2)$ and $p(\mathbf{w}) \propto \exp(-\frac{\alpha}{2}\|\mathbf{w}\|^2)$. Taking $-\log$ and dividing by β, maximising the posterior is minimising

$$\tilde{E}(\mathbf{w}) = \tfrac{1}{2}\|\mathbf{\Phi w} - \mathbf{t}\|^2 + \tfrac{\lambda}{2}\|\mathbf{w}\|^2, \qquad \lambda = \alpha/\beta.$$

Setting $\nabla\tilde{E} = \mathbf{\Phi}^\top(\mathbf{\Phi}\mathbf{w} - \mathbf{t}) + \lambda\mathbf{w} = \mathbf{0}$ gives $(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)\mathbf{w} = \mathbf{\Phi}^\top\mathbf{t}$. For $\lambda > 0$ the matrix is symmetric positive definite, so the solution is unique; we solve the system directly for every λ in the grid (49 values from 10^{-10} to 10^{2}), in the $d = 9$ monomial basis of P1.

GD at $\lambda = 10^{-2}$. Plain batch GD in the monomial basis, $\mathbf{w} \leftarrow \mathbf{w} - \eta\,(\mathbf{\Phi}^\top(\mathbf{\Phi}\mathbf{w} - \mathbf{t}) + \lambda\mathbf{w})$ from $\mathbf{w} = \mathbf{0}$ with $\eta = 0.04$, reached $\|\mathbf{w}_{\mathrm{GD}} - \mathbf{w}_{\mathrm{closed}}\|/\|\mathbf{w}_{\mathrm{closed}}\| < 10^{-6}$ after 28,227 iterations (Figure 4); its final objective 0.4840278 matches the closed form's 0.4840278. Here $\mathrm{cond}(\mathbf{\Phi}^\top\mathbf{\Phi}) = 2.31 \times 10^{14}$ and $\mathrm{cond}(\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I) = 2038$: the penalty adds λ to every eigenvalue, lifting the smallest from almost zero to about λ and cutting the condition number by eleven orders of magnitude, which is what makes plain GD in this basis practical.

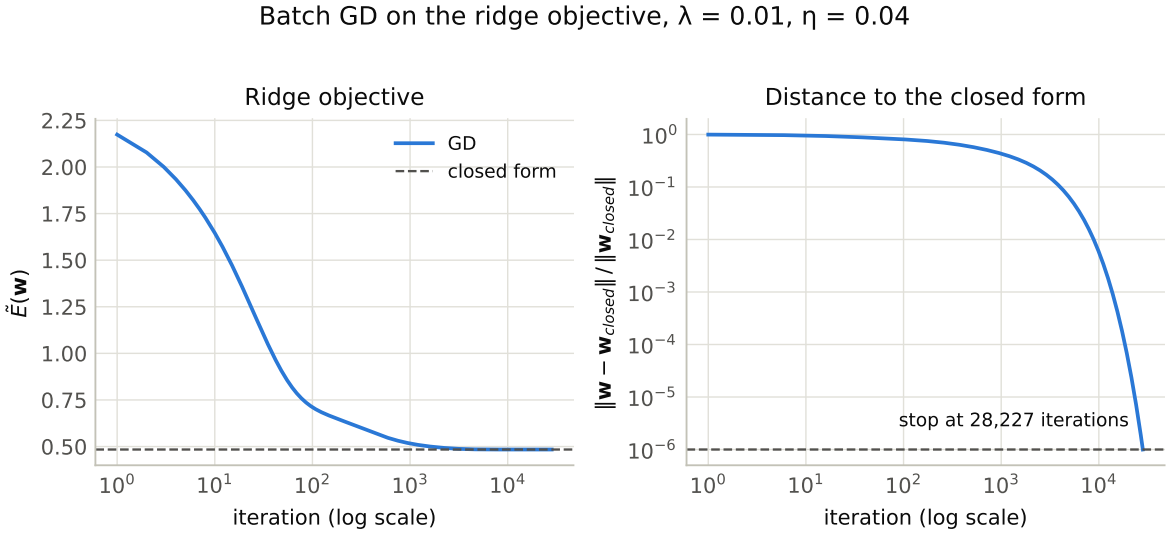


Figure 4: Batch GD on the ridge objective at $\lambda = 10^{-2}$: $\tilde{E}(\mathbf{w})$ per iteration (left) and the relative distance to the closed-form solution (right).

RMSE sweep and $\lambda^\star$. Figure 5 shows train and test RMSE over the grid. The test RMSE is smallest at $\lambda^\star = 1.78 \times 10^{-7}$, where it is 0.1146 (train 0.05410). It is within 1.05× its minimum on the 8 contiguous grid points $\lambda \in [3.16 \times 10^{-8}, 1.78 \times 10^{-6}]$ around $\lambda^\star$, the interval; the qualifying set is not one interval, though, because after rising to at most 1.066× the minimum (at $\lambda = 10^{-5}$) the test RMSE dips back under 1.05× at the isolated grid point $\lambda = 10^{-4}$. The minimum is flat: the test RMSE stays within 5% of its minimum over almost two decades of λ around $\lambda^\star$ and never exceeds 1.066× it from 3.16×10^{-8} to 10^{-4}, so the exact choice of λ in this range hardly matters. Below the interval the penalty is too weak and the degree-9 polynomial chases the noise (at $\lambda = 10^{-10}$ the train RMSE falls to 0.01649 while the test RMSE climbs to 0.1605), and above it the test RMSE first plateaus near 0.12 until $\lambda \approx 10^{-4}$, after which the over-shrunk weights underfit and train and test RMSE rise together (to 0.2660 at $\lambda = 10^{-2}$).

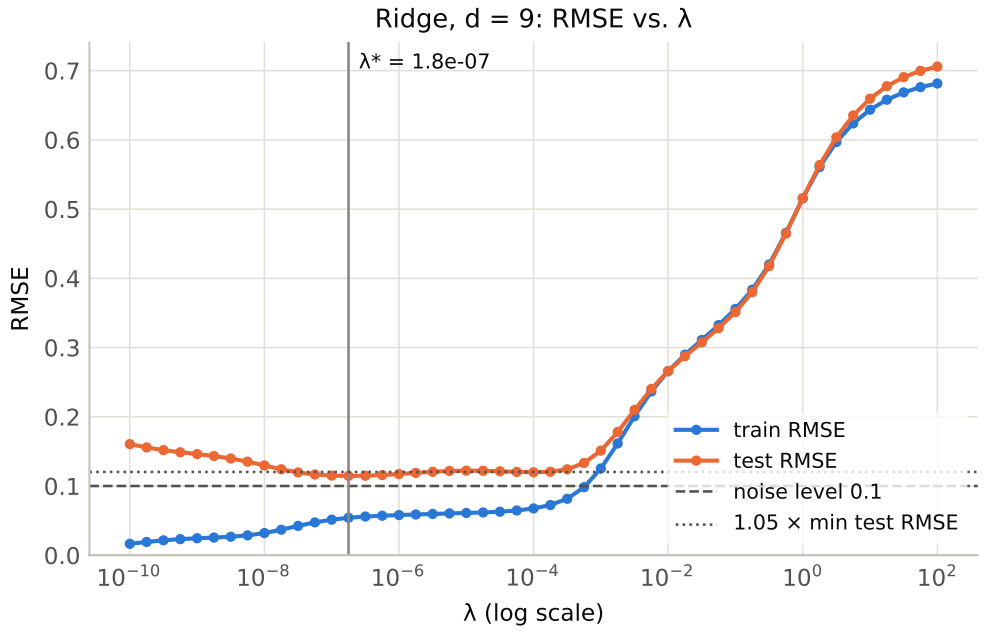


Figure 5: Train and test RMSE of ridge regression ($d = 9$) vs. λ, with the noise level $\sqrt{0.01} = 0.1$ (dashed) and 1.05× the minimum test RMSE (dotted).

Fitted curves. Figure 6 shows the fits at the three values of λ, and Table 4 their weight norms. The norm falls by almost three orders of magnitude from the overfit to the underfit model: the penalty buys smoothness by forbidding the large, cancelling coefficients an interpolating polynomial needs.

	λ	$\|\mathbf{w}\|_2$	train RMSE	test RMSE
overfit	10^{-10}	3979	0.01649	0.1605
$\lambda^\star$	1.78×10^{-7}	124	0.05410	0.1146
underfit	10^{-2}	5.11	0.2659	0.2660

Table 4: The three ridge fits of Figure 6.

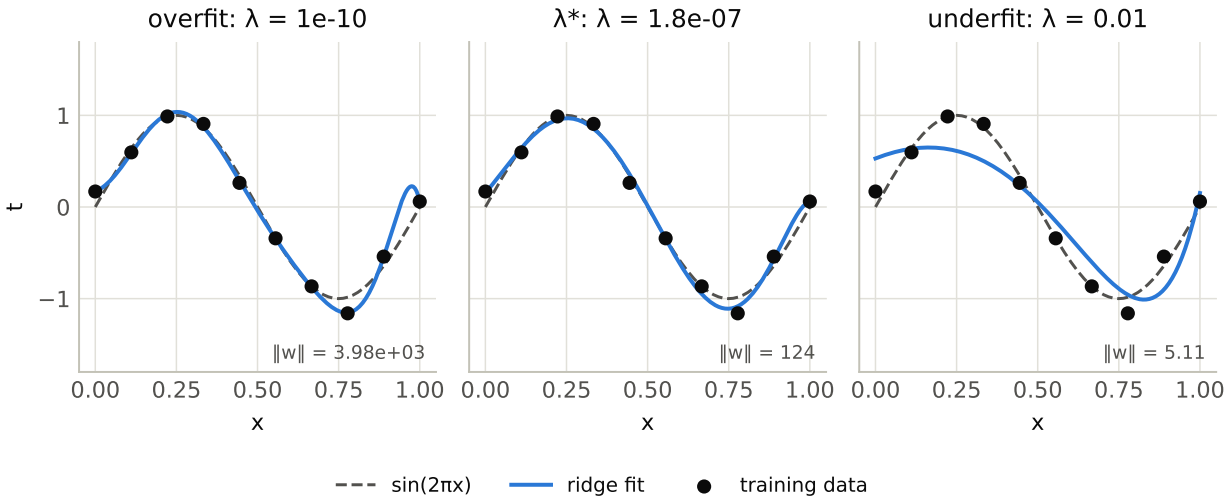


Figure 6: Ridge fits ($d = 9$) for $\lambda = 10^{-10}$, for $\lambda^\star$ and for $\lambda = 10^{-2}$, with the training data and $\sin(2\pi x)$.

3 P3: L1-regularised regression via subgradient descent

Method. With $u = 2x - 1$, the design matrix is $\Phi_{nj} = P_j(u_n)$, $j = 0, \ldots, 9$ (NumPy's `legvander` applied to $2x - 1$). It spans the same degree-9 polynomials as the monomial basis, but $\mathrm{cond}(\mathbf{\Phi}^\top \mathbf{\Phi}) = 611$ instead of about 10^{14}. Subgradient descent on $E(\mathbf{w}) = \frac{1}{2}\|\mathbf{\Phi w} - \mathbf{t}\|^2 + \lambda\|\mathbf{w}\|_1$ runs $\mathbf{w} \leftarrow \mathbf{w} - \eta\,(\mathbf{\Phi}^\top(\mathbf{\Phi w} - \mathbf{t}) + \lambda\,\mathrm{sign}(\mathbf{w}))$ with $\mathrm{sign}(0) = 0$ and $\eta = 0.05$ from $\mathbf{w} = \mathbf{0}$ for 20,000 iterations at each of the 21 values of λ from 10^{-4} to 1. Every reported quantity comes from the final iterate, and a coefficient counts as ≈ 0 when $|w_j| < 2\eta\lambda$, since with a constant step a zero coefficient chatters around 0 with amplitude about $\eta\lambda$.

Convergence and verification at $\lambda = 0.1$. Figure 7 shows E and the best E so far. Subgradient descent is not a descent method: once near the optimum, the sign term flips for the coefficients at zero and E rises and falls by about 10^{-3} (right panel), while the best-so-far E keeps creeping down. The final iterate has $E = 0.2636$, against the reference $E^\star = 0.2625$, and

$$\mathbf{w} = (-0.006,\ -0.930,\ 0.043,\ 1.115,\ 0.030,\ -0.203,\ -0.002,\ -0.025,\ 0.002,\ 0.001),$$
$$\mathbf{w}^\star = (0.000,\ -0.928,\ 0.045,\ 1.117,\ 0.032,\ -0.201,\ 0.000,\ -0.026,\ 0.000,\ 0.000),$$

so $\|\mathbf{w} - \mathbf{w}^\star\|_2 = 0.00764$, the size of the $\eta\lambda$ chatter. The optimality conditions hold to within that chatter: with $\mathbf{r} = \mathbf{t} - \mathbf{\Phi w}$, over the 6 coefficients that are not ≈ 0, $\max_j |\mathbf{\Phi}_j^\top \mathbf{r} - \lambda\,\mathrm{sign}(w_j)| = 0.0144$, and over the 4 that are, $\max_j |\mathbf{\Phi}_j^\top \mathbf{r}|/\lambda = 0.683 \le 1$.

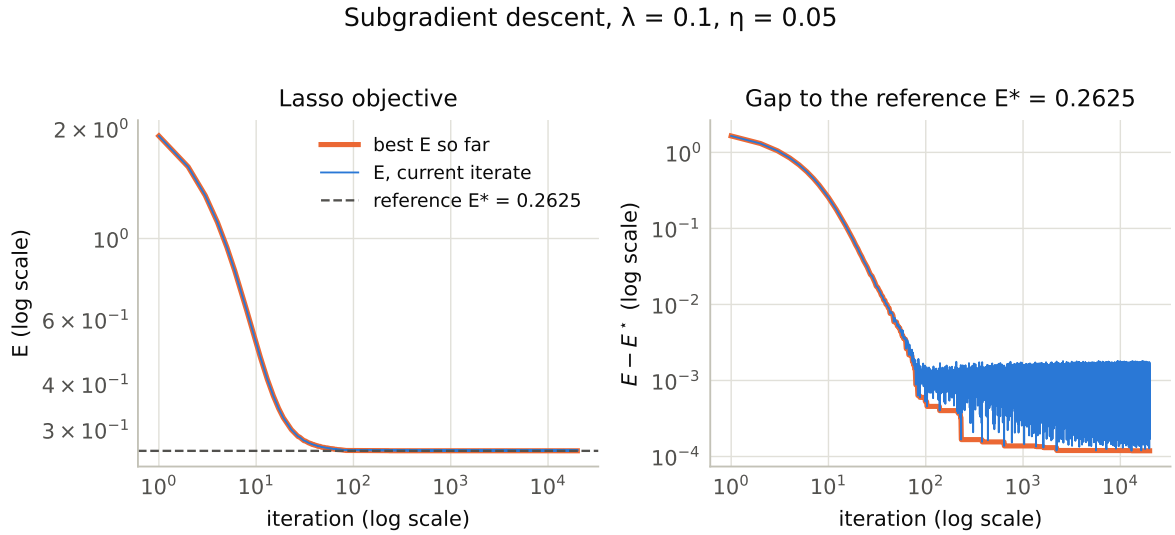


Figure 7: Subgradient descent at $\lambda = 0.1$: E and the best E so far (left), and both minus the reference $E^\star$ (right). The iteration axis is logarithmic so that the fast first 100 iterations are visible.

RMSE sweep and $\lambda^\star$. Figure 8 shows train and test RMSE. The test RMSE is smallest at $\lambda^\star = 10^{-1}$, where it is 0.09854; there 4 coefficients are ≈ 0, with indices $j \in \{0, 6, 8, 9\}$, exactly the zeros of $\mathbf{w}^\star$.

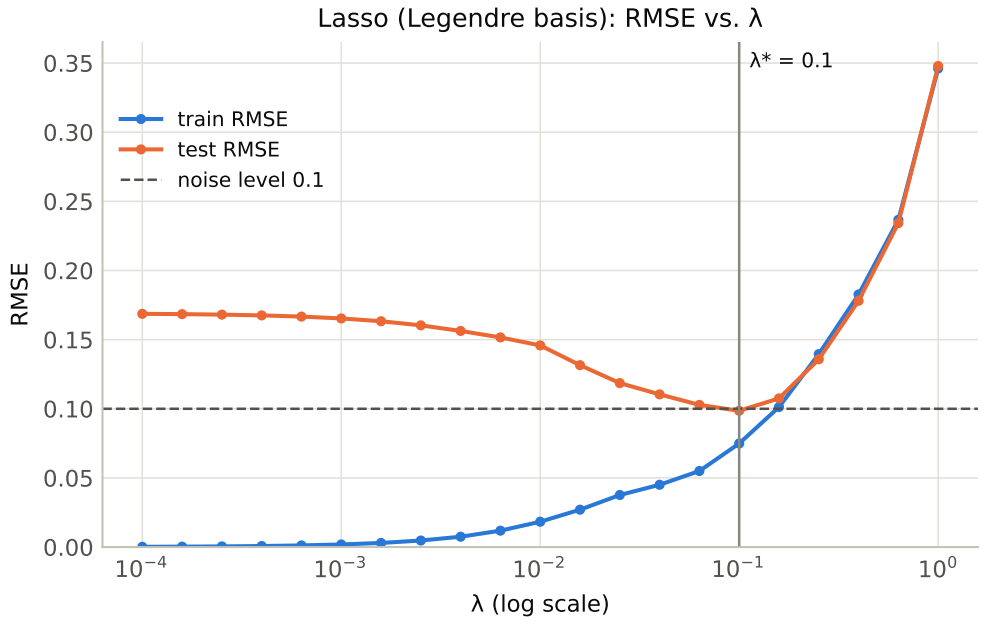


Figure 8: Train and test RMSE of the Lasso (Legendre basis) vs. λ, with the noise level 0.1.

Coefficient path. Figure 9 shows every w_j against λ, and Table 5 how many are ≈ 0 at each λ.

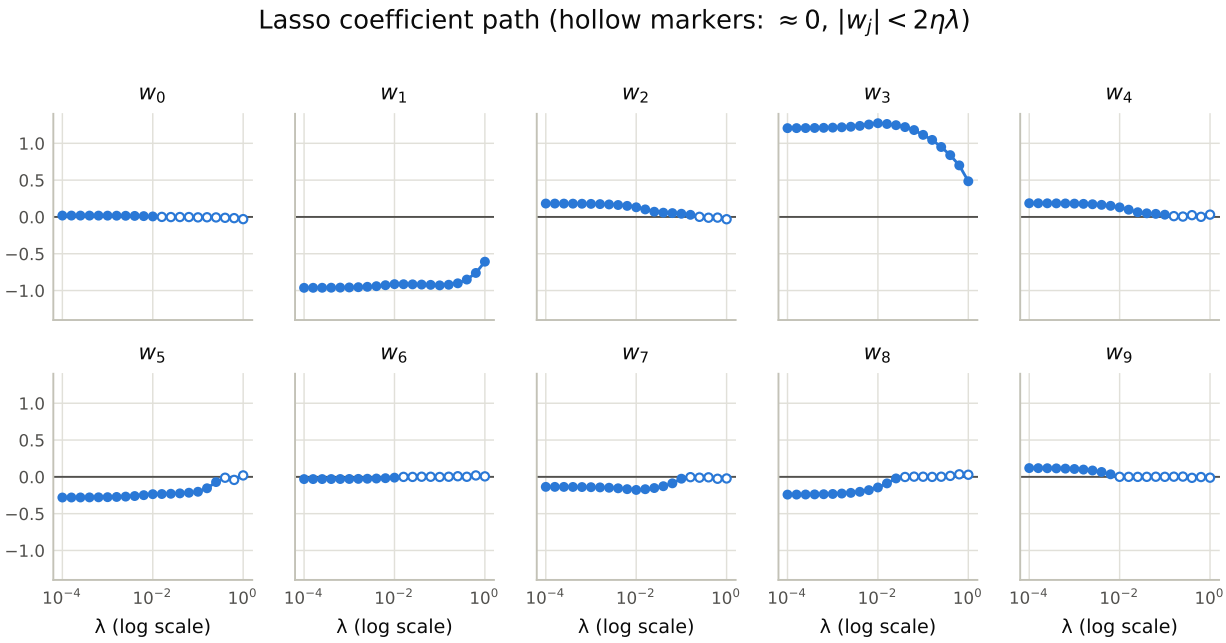


Figure 9: Lasso coefficient path; hollow markers are coefficients counted as ≈ 0 ($|w_j| < 2\eta\lambda$).

λ	$\#\approx 0$	indices j	train RMSE	test RMSE
10^{-4}	0	–	1.88×10^{-4}	0.169
1.58×10^{-4}	0	–	2.98×10^{-4}	0.168
2.51×10^{-4}	0	–	4.72×10^{-4}	0.168
3.98×10^{-4}	0	–	7.48×10^{-4}	0.168
6.31×10^{-4}	0	–	0.00119	0.167
10^{-3}	0	–	0.00188	0.165
1.58×10^{-3}	0	–	0.00298	0.163
2.51×10^{-3}	0	–	0.00472	0.160
3.98×10^{-3}	0	–	0.00748	0.156
6.31×10^{-3}	0	–	0.0119	0.152
10^{-2}	1	9	0.0183	0.146
1.58×10^{-2}	3	0, 6, 9	0.0270	0.132
2.51×10^{-2}	3	0, 6, 9	0.0376	0.119
3.98×10^{-2}	4	0, 6, 8, 9	0.0450	0.110
6.31×10^{-2}	4	0, 6, 8, 9	0.0550	0.103
10^{-1}	4	0, 6, 8, 9	0.0749	0.0985
1.58×10^{-1}	6	0, 4, 6, 7, 8, 9	0.101	0.107
2.51×10^{-1}	7	0, 2, 4, 6, 7, 8, 9	0.140	0.136
3.98×10^{-1}	8	0, 2, 4, 5, 6, 7, 8, 9	0.183	0.178
6.31×10^{-1}	8	0, 2, 4, 5, 6, 7, 8, 9	0.237	0.234
10^{0}	8	0, 2, 4, 5, 6, 7, 8, 9	0.346	0.348

Table 5: Coefficients counted as ≈ 0 ($|w_j| < 2\eta\lambda$) at each λ.

Lasso vs. Ridge. Ridge in the same Legendre basis, solved in closed form on the same grid, has its best test RMSE at $\lambda = 1.58 \times 10^{-1}$:

	best λ	test RMSE	$\|\mathbf{w}\|_2$	$\#\ |w_j| < 0.01$
Lasso	10^{-1}	0.09854	1.47	4
Ridge	1.58×10^{-1}	0.1422	1.44	1

Table 6: Lasso and Ridge, each at its own best λ (Legendre basis, $d = 9$).

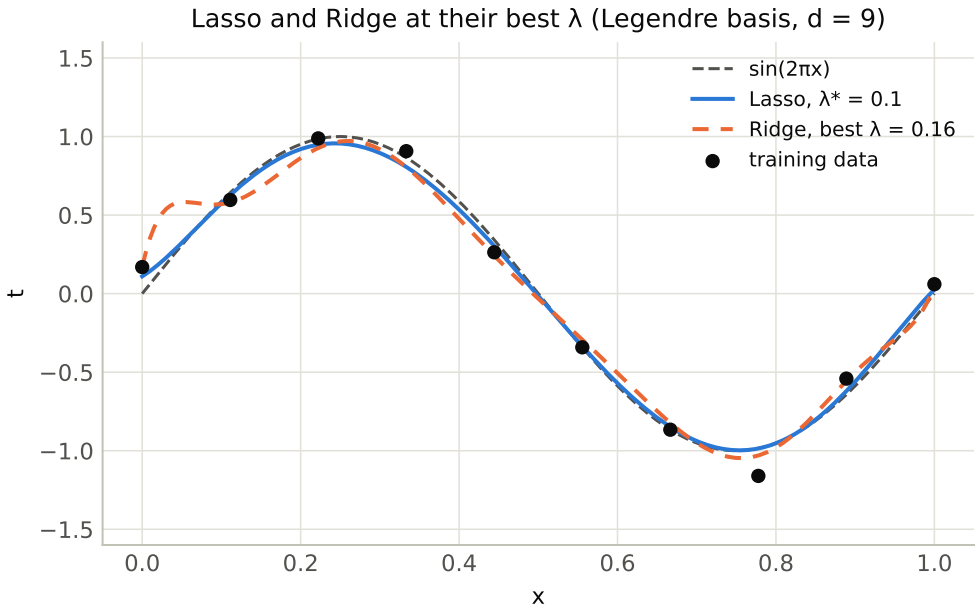


Figure 10: The Lasso fit at $\lambda^\star$ and the Ridge fit at its own best λ, with the training data and $\sin(2\pi x)$.

Discussion. L1 produces exact zeros because the subgradient of $|w_j|$ at 0 is the whole interval $[-1, 1]$: $w_j = 0$ is optimal whenever $|\boldsymbol{\Phi}_j^\top \mathbf{r}| \le \lambda$, so a feature whose correlation with the residual stays below λ is switched off entirely, and as λ grows the coefficients cross that threshold one after another (Figure 9, Table 5). The L2 gradient λw_j vanishes at 0, so Ridge pulls each coefficient toward zero in proportion to its size and never gets there; geometrically, the SSE contours meet the ℓ_1 ball at a corner on an axis but the round ℓ_2 ball away from the axes. Table 6 shows the result: at almost the same $\|\mathbf{w}\|_2$, the Lasso has 4 coefficients below 0.01 and Ridge only 1. The first coefficients to go carry almost no signal ($\sin(2\pi x) = -\sin(\pi u)$ has sizeable Legendre coefficients only on P_1, P_3 and P_5), so up to $\lambda^\star$ the Lasso mostly removes noise and its test RMSE falls; past $\lambda^\star$ it shrinks the signal coefficients w_1, w_3 and w_5, and train and test RMSE rise together as bias takes over.

4 P4: Bayesian regression

Method. With $\boldsymbol{\phi}(x) = [1, x, x^2, x^3]^\top$, prior $\mathcal{N}(\mathbf{0}, \alpha^{-1} I)$ and noise precision β, the posterior is $\mathcal{N}(\mathbf{w} \mid \mathbf{m}_N, S_N)$ with $S_N^{-1} = \alpha I + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi}$ and $\mathbf{m}_N = \beta S_N \boldsymbol{\Phi}^\top \mathbf{t}$, and the predictive distribution is $\mathcal{N}(m(x), s^2(x))$ with $m(x) = \boldsymbol{\phi}(x)^\top \mathbf{m}_N$ and $s^2(x) = \beta^{-1} + \boldsymbol{\phi}(x)^\top S_N \boldsymbol{\phi}(x)$. We use $\alpha = 0.05$ and $\beta = 1/0.01 = 100$ for every N. S_N is never formed: with $A = S_N^{-1}$, both $\mathbf{m}_N = \beta A^{-1} \boldsymbol{\Phi}^\top \mathbf{t}$ and $\boldsymbol{\phi}^\top S_N \boldsymbol{\phi} = \boldsymbol{\phi}^\top A^{-1} \boldsymbol{\phi}$ come from linear solves with A.

Equivalence. Since $\mathbf{m}_N = \beta(\alpha I + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^\top \mathbf{t} = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \frac{\alpha}{\beta} I)^{-1} \boldsymbol{\Phi}^\top \mathbf{t}$, the posterior mean is exactly the ridge solution with $\lambda = \alpha/\beta = 0.0005$; a Gaussian posterior's mean is also its mode, so MAP and posterior mean coincide, and so do $\mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}(x)$ and $m(x)$. Numerically, for $N = 10$, $\mathbf{m}_N = (0.09384,\ 7.368,\ -22.81,\ 15.25)$, $\mathbf{w}_{\text{MAP}} = (0.09384,\ 7.368,\ -22.81,\ 15.25)$ and $\|\mathbf{w}_{\text{MAP}} - \mathbf{m}_N\|_2 = 1.0 \times 10^{-12}$: the two differ only by rounding, and their curves coincide in Figure 11.

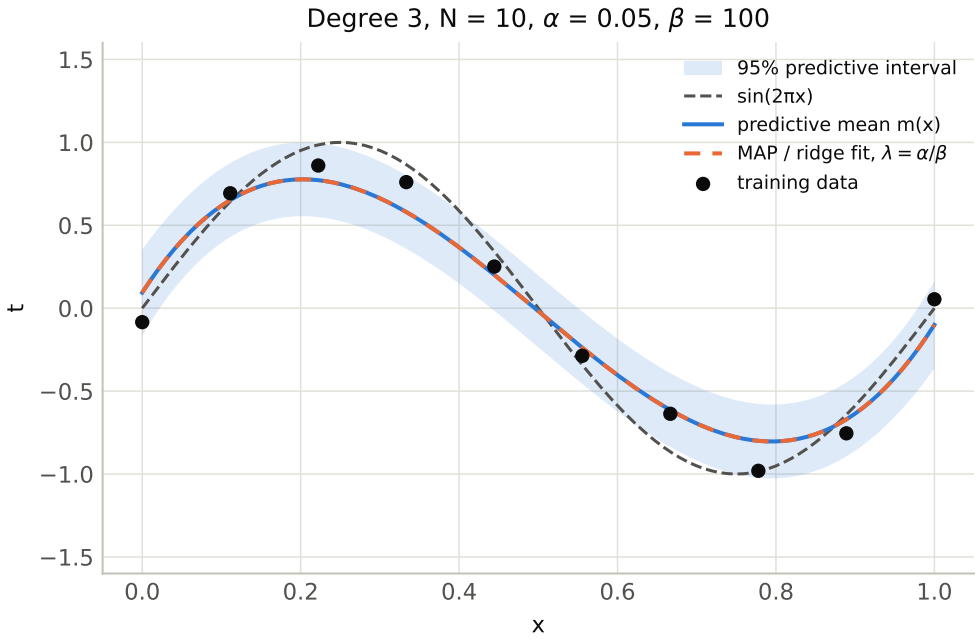


Figure 11: Posterior predictive mean and 95% interval $m(x) \pm 1.96\, s(x)$ for $N = 10$, with the MAP (ridge) fit overlaid, the training data and $\sin(2\pi x)$.

Effect of N. Figure 12 shows the interval width $2 \times 1.96\, s(x)$ for $N = 5, 10, 20, 50$:

N	mean width	largest width	parameter share of s^2 (%)
5	0.495	0.549	37
10	0.449	0.524	24
20	0.424	0.486	14
50	0.406	0.442	6.8

Table 7: Predictive interval width over $x \in [0, 1]$, and the average share of the predictive variance that is parameter uncertainty $\boldsymbol{\phi}^\top S_N \boldsymbol{\phi}$. The noise floor is $2 \times 1.96\, \beta^{-1/2} = 0.392$.

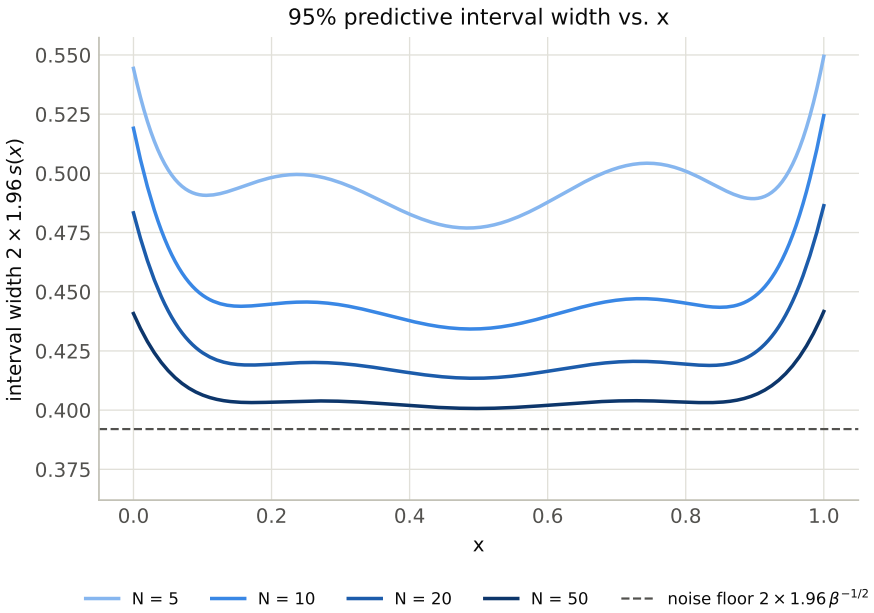


Figure 12: Width of the 95% predictive interval vs. x for each N, with the irreducible-noise floor.

Discussion. Every training point adds $\beta \boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^\top$ to the posterior precision S_N^{-1}, so the posterior covariance S_N, and with it the parameter uncertainty $\boldsymbol{\phi}(x)^\top S_N \boldsymbol{\phi}(x)$, shrinks as N grows: roughly like $1/N$ in the middle of $[0, 1]$, more slowly at the ends, where a cubic is least pinned down and the band stays widest (its share of the predictive variance falls from 37% at $N = 5$ to 6.8% at $N = 50$). The observation-noise term β^{-1} does not depend on the data at all: it is the scatter of a new target around the true function, so no amount of data removes it, and the band can only approach the floor $2 \times 1.96\, \beta^{-1/2} = 0.392$, never cross it.